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Some Remarks on Finite Element Approximations of Crack Problems and an Analysis of Hybrid Methods

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Some Remarks on Finite Element Approximations of Crack Problems and an Analysis of Hybrid Methods

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ABSTRACT

We discuss here several results on the convergence of finite element methods for problems with corners and cracks. We also derive some *a priori* estimates for a certain class of hybrid finite elements.

I. INTRODUCTION

The use of finite elements in problems involving singularities has generated considerable interest in recent years due to the importance of singular behavior in fracture mechanics. Surveys of the literature on this subject, together with descriptions of a multitude of finite element methods for crack problems, have been presented by Gallagher [1] and Pian [2]. Studies of the behavior of finite element solutions in domains with cracks and reentrant corners have been made by Babuska [3], Babuska and Rosenwieg [4], Barnhill and Whiteman [5], and, more recently, by Schatz and Whalbin [6].

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In the present paper we review some of the principal results on the mathematical aspects of convergence of finite element methods for crack problems which should be of particular interest to practitioners, and we describe briefly some results on a type of hybrid crack element.

II. SOME DIFFICULTIES WITH STANDARD ELEMENTS

An important component of any mathematical theory on the convergence of finite element methods is an associated theory of interpolation; i.e., given a function u of a specified degree of smoothness, how well can it be approximated using a given finite element mesh? This question can actually be regarded as being independent of questions of approximation of solutions of differential equations, but is obviously an essential prerequisite to a viable approximation theory.

One of the most important results on the interpolation theory of finite elements was contributed by Ciarlet and Raviart [7] (further generalizations were made by Ciarlet [8, 9]). Let Ω be a domain in $\mathbf{R}^n$ satisfying a cone condition (see, for example, Oden and Reddy [10]) and let $\mathcal{Q}_h$ be a partition of Ω into finite elements; i.e., into subdomains $\{\Omega_e\}_{e \in I}$, $\bar{\Omega} = \bigcup_{e \in I} \bar{\Omega}_e$, $\Omega_e \cap \Omega_f = \emptyset$, $e \neq f$. Let each element Ω_e be the image of a fixed master element $\bar{\Omega}$ under an affine invertible map T_e . We assume that within each element the local approximations (i.e., the local interpolation of functions) contain complete polynomials of degree k , $k \geq 1$. Further let

$$h_e = \text{diameter}(\Omega_e) \text{ and } \rho_e = \sup \{ \text{diameters of all spheres contained in } \Omega_e \}$$

Suppose $u \in W^{k+1,p}(\Omega_e)$, $1 \leq p \leq \infty$, and Π_{Ω_e} is a projection of $W^{k+1,p}(\Omega_e)$ onto $W^{m,p}(\Omega_e)$, $0 \leq m \leq k$, which preserves polynomials of degree $\leq k$. Then Ciarlet and Raviart [7] have shown that a positive constant C , independent of h_e and u , exists such that

$$\|u - \Pi_{\Omega_e} u\|_{m,p,\Omega_e} \leq C \frac{h_e^{k+1}}{\rho_e^m} \|u\|_{k+1,p,\Omega_e} \quad (1)$$

Here we use the usual notations for norms on Sobolev spaces:

$$\begin{aligned} \|u\|_{m,p,\Omega} &= \left\{ \int_{\Omega} \sum_{|\alpha| \leq m} |D^\alpha u|^p dx \right\}^{1/p}, \quad 1 \leq p < \infty \\ \|u\|_{m,2,\Omega} &\equiv \|u\|_{m,\Omega} \\ \|u\|_{m,\infty,\Omega} &= \sup_{x \in \Omega} \max_{0 \leq |\alpha| \leq m} |D^\alpha u(x)| \end{aligned} \quad (2)$$

Here

$$\begin{aligned} \alpha &= (\alpha_1, \alpha_2, \dots, \alpha_n), \quad \alpha_i = \text{integer} \geq 0, |\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_n \\ D^\alpha u &= \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \dots \partial x_n^{\alpha_n}}, \quad \mathbf{x} = (x_1, x_2, \dots, x_n) \in \Omega \subset \mathbf{R}^n \\ dx &= dx_1 dx_2 \dots dx_n \end{aligned} \quad (3)$$

If the mesh refinements are regular; i.e., if there exists a constant $\sigma_0 > 0$ such that $h_e/\rho_e \leq \sigma_0$ as $h_e \rightarrow 0$, and if $h = \max_{1 \leq e \in E} (h_e)$, then (1) yields the following global estimate: There exists a $C > 0$ and an element $\phi_h \in S_h(\Omega)$ such that

$$\|u - \phi_h\|_{m,p,\Omega} \leq Ch^{k+1-m} \|u\|_{k+1,p,\Omega} \quad (4)$$

Here $S_h(\Omega)$ is the finite-dimensional subspace of $W^{m,p}(\Omega)$ spanned by the global finite-element basis functions. Thus, by appropriate refinements of the finite element mesh, we produce a family of subspaces $\{S_h(\Omega)\}_{0 < h \leq 1}$. The union of all of these subspaces (as h tends to zero) is everywhere dense in $W^{m,p}(\Omega)$ and the rate at which the interpolants $\phi_h (= \Pi_h u)$ converge to u in $W^{m,p}(\Omega)$ is h^{k+1-m} provided u is at least in $W^{k+1,p}(\Omega)$.

The interpolation estimate (2) has immediate application to the estimation of errors in finite element approximations of boundary-value problems. For example, consider the general elliptic boundary-value problem of order $2m$, $m > 0$, characterized by

$$\begin{aligned} Au &= f \text{ in } \Omega \\ \gamma_j u &= 0 \text{ on } \partial\Omega, \quad 0 \leq j \leq m-1 \end{aligned} \quad (5)$$

where A is the general elliptic operator

$$Au \equiv \sum_{|\alpha|+|\beta| \leq m} (-1)^{|\alpha|} D^\alpha a_{\alpha\beta}(\mathbf{x}) D^\beta u(\mathbf{x}) \quad (6)$$

and γ_j denote the traces (normal derivatives) of u on the boundary $\partial\Omega$ of an open bounded domain Ω in $\mathbf{R}^n$. In (3), $a_{\alpha\beta}$ are variable coefficients, $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbf{R}^n$, and the multi-index notation of (3) is used.

Instead of (2), we seek solutions to the equivalent variational boundary-value problem: Find $u \in H_0^m(\Omega)$ such that

$$B(u, v) = f(v) \quad \forall v \in H_0^m(\Omega) \quad (7)$$

where

$$\begin{aligned} B(u, v) &= \int_{\Omega} \sum_{|\alpha|, |\beta| \leq m} a_{\alpha\beta} D^{\alpha} u D^{\beta} v \, dx \\ f(v) &= \int_{\Omega} f v \, dx \end{aligned} \quad (8)$$

Here $H^m(\Omega) \equiv W^{m,2}(\Omega)$ and $H_0^m(\Omega)$ is the closure of $C_0^{\infty}(\Omega)$ in $H^m(\Omega)$.

The Galerkin approximation of (7) consists of seeking a function u_h belonging to a finite-dimensional subspace $S_h(\Omega)$ of $H_0^m(\Omega)$ such that

$$B(u_h, v_h) = f(v_h) \quad \forall v_h \in S_h(\Omega) \quad (9)$$

It is well known that if the following conditions hold, a unique solution exists to (7) (see, e.g., Ref. 10):

1. There exists a constant $M > 0$ such that

$$B(u, v) \leq M \|u\|_{m,\Omega} \|v\|_{m,\Omega} \quad \forall u, v \in H_0^m(\Omega)$$

2. There exists a constant γ such that

$$\inf_{\|u\|_{m,\Omega}=1} \sup_{\|v\|_{m,\Omega} \leq 1} |B(u, v)| \geq \gamma > 0 \quad (10)$$

- 3.

$$\sup_{u \in H_0^m(\Omega)} |B(u, v)| > 0, \quad v \neq 0$$

Similarly, (9) possesses a unique solution if (10) hold on the subspace $S_h(\Omega)$, with γ in Condition 2 possibly replaced by a positive constant γ_h dependent on the parameter h .

Under these conditions, it can be shown (e.g., Ref. 10) that the *approximation error*

$$e_h = u - u_h \quad (11)$$

satisfies

$$\|e_h\|_{m,\Omega} \leq \left(1 + \frac{M}{\gamma_h}\right) \|u - v_h\|_{m,\Omega} \quad \forall v_h \in S_h(\Omega) \quad (12)$$

Hence, by selecting v_h to be the interpolant ϕ_h in (4), we arrive at the error

estimate

$$\|e_h\|_{m,\Omega} \leq C \left(1 + \frac{M}{\gamma_h}\right) h^{k+1-m} \|u\|_{k+1,\Omega} \quad (13)$$

Unfortunately, (13) holds only if u is sufficiently regular. If $u \in H^r(\Omega) \cap H_0^m(\Omega)$, (13) is replaced by the estimate

$$\begin{aligned} \|e_h\|_{m,\Omega} &\leq C_0 h^\mu \|u\|_{r,\Omega} \\ \mu &= \min(k+1-m, r-m) \end{aligned} \quad (14)$$

where C_0 is a constant. Thus if $r < k+1$, the global rate of convergence is not affected by the degree k of the polynomials used in the finite element interpolation.

The latter observation is particularly important in applications involving irregular domains. The regularity of the solution of an elliptic boundary-value problem is governed by the smoothness of the coefficients appearing in the operators defining the problem, the smoothness of the data, and the regularity of the domain Ω on which the problem is defined. Even if the coefficients and data are smooth, irregularities such as cracks or reentrant corners can adversely affect the global regularity of the solution and, therefore, degrade the convergence rates of finite element approximations.

To emphasize this point, consider the case of a plane polygonal domain $\Omega \subset \mathbb{R}^2$ with a boundary $\partial\Omega$ consisting of M straight-line segments (or smooth arcs) meeting at vertices P_j , $j = 1, 2, \dots, M$, having interior angles $0 < \alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_M < 2\pi$ ordered in such a way that $\max_{1 \leq j \leq M} \alpha_j = \alpha_M$ (see Fig. 1). Suppose that we wish to solve Laplace's equation on this domain; i.e.,

$$\begin{aligned} -\Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \end{aligned} \quad (15)$$

where $\Delta u = u_{xx} + u_{yy}$ and $f = f(x, y)$ is smooth (e.g., $f \in C^\infty(\bar{\Omega})$). Then if

$$\beta_j = \pi/\alpha_j \quad (16)$$

the solution u of (15) is just smooth enough to satisfy

$$u \in H^{1+\beta_M-\varepsilon}(\Omega) \quad (17)$$

where ε is an arbitrary positive number. Thus for the domains indicated in

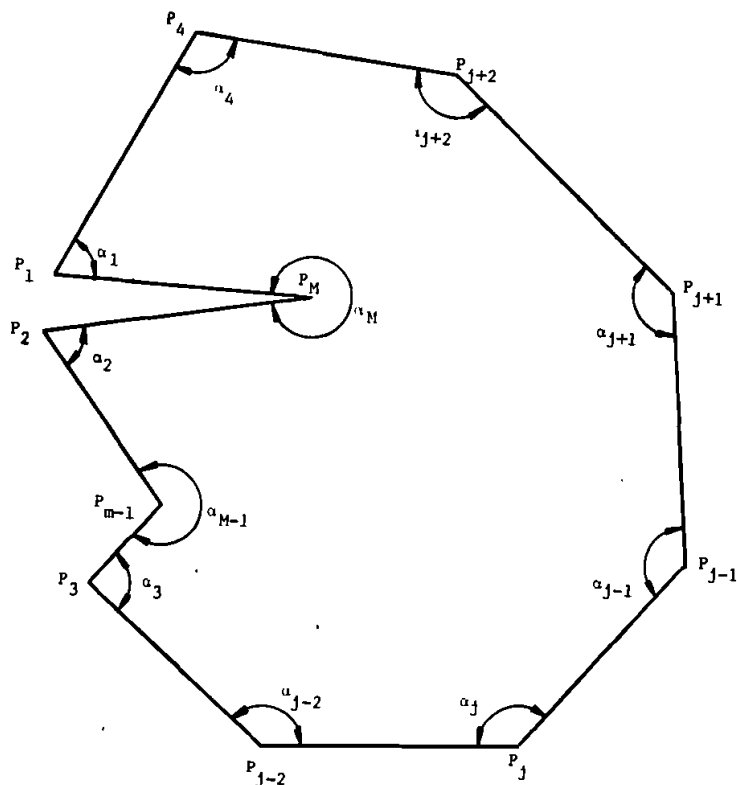


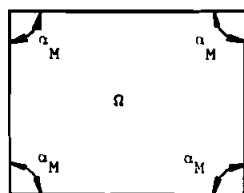
Fig. 1 Polygonal domain $\Omega \subset \mathbb{R}^2$.

Fig. 2 we have

$$\begin{aligned}
 \text{Rectangular: } & u \in H^{3-\epsilon}(\Omega) \cap H_0^1(\Omega) \\
 \text{L-shaped: } & u \in H^{5/3-\epsilon}(\Omega) \cap H_0^1(\Omega) \\
 \text{Cracked: } & u \in H^{3/2-\epsilon}(\Omega) \cap H_0^1(\Omega)
 \end{aligned} \tag{18}$$

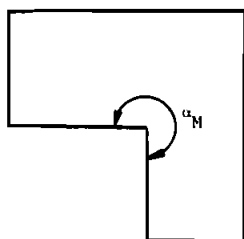
Thus if polynomials of degree k are employed in the finite element approximation of (15), the global rates of convergence in $H^1(\Omega)$ for the problems in Fig. 2 are

$$\begin{aligned}
 \text{Rectangular: } & \|u - \phi_h\|_{1,\Omega} \leq Ch^{\min(k, 2-\epsilon)} \|u\|_{3-\epsilon,\Omega} \\
 \text{L-shaped: } & \|u - \phi_h\|_{1,\Omega} \leq Ch^{2/3-\epsilon} \|u\|_{5/3-\Omega} \\
 \text{Cracked: } & \|u - \phi_h\|_{1,\Omega} \leq Ch^{1/2-\epsilon} \|u\|_{3/2,\Omega}
 \end{aligned} \tag{19}$$



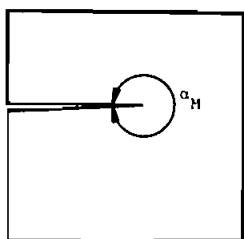
$$\alpha_M = \frac{\pi}{2}, \quad \beta_M = 2$$

Rectangular



$$\alpha_M = \frac{3\pi}{2}, \quad \beta_M = \frac{2}{3}$$

L-shaped



$$\alpha_M = 2\pi - \epsilon, \quad \beta_M = \frac{1}{2} + \epsilon'$$

Cracked

Fig. 2 Some common irregular domains.

Thus, in all of these cases except the rectangular domain, the rate of convergence in $H^1(\Omega)$ is not influenced at all by the order of the polynomial k , and, even in the rectangular case, only the choice $k = 1$ controls the rate of convergence.

III. LOCAL ESTIMATES

Estimates such as (19) reflect the worst behavior of the solution on the entire domain Ω . Locally, the situation may be much better. Let Ω_j denote the intersection of Ω with a collection of open balls centered at each vertex with radii small enough that each ball contains only one vertex and denote $\Omega_0 = \Omega - \bigcup_{j=1}^M \Omega_j$ (see Fig. 3). Then locally

$$\begin{aligned} \|u - \phi_h\|_{1,\Omega_j} &\leq Ch^{\min(k,\beta_j)-\varepsilon} \|u\|_{\beta_j+1-\varepsilon,\Omega_j} \\ \|u - \phi_h\|_{1,\Omega_0} &\leq Ch^k \|u\|_{k+1,\Omega_0} \end{aligned} \quad (20)$$

These estimates are given in Schatz and Whalbin [6].

Schatz and Whalbin [6] obtained some of the most important results to date on local interior estimates for polygonal domains. Let Ω_0 and $\hat{\Omega}$ be interior subdomains of Ω such that $\Omega_0 \subset \subset \hat{\Omega} \subset \subset \Omega$; i.e., the closure of $\hat{\Omega}$ is strictly interior to Ω and the closure of Ω_0 is strictly interior to $\hat{\Omega}$ (by "strictly interior" we mean that $\text{dist}(\hat{\Omega}, \partial\Omega) > 0$, etc.). Then it is shown in Ref. 6 that for any $v_h \in S_h(\Omega)$,

$$\|u - u_h\|_{\infty,\Omega_0} \leq C \left[\left(\ln \frac{1}{h} \right)^\gamma \|u - v_h\|_{\infty,\hat{\Omega}} + \|u - u_h\|_{-s,\hat{\Omega}} \right] \quad (21)$$

wherein

$$\gamma = \begin{cases} 1 & \text{if } k = 1 \\ 0 & \text{if } k \geq 2 \end{cases}$$

and $\|\cdot\|_{-s,\hat{\Omega}}$ denotes the norm on the dual space $(H_0^s(\Omega))'$ for s an arbitrary nonnegative integer. Locally, it can be shown that

$$\|u - v_h\|_{\infty,\hat{\Omega}} \leq Ch^{k+1} \|u\|_{k+1,\hat{\Omega}} \quad (22)$$

The term $\|u - u_h\|_{-s,\hat{\Omega}}$ reflects the influence on the local error of factors

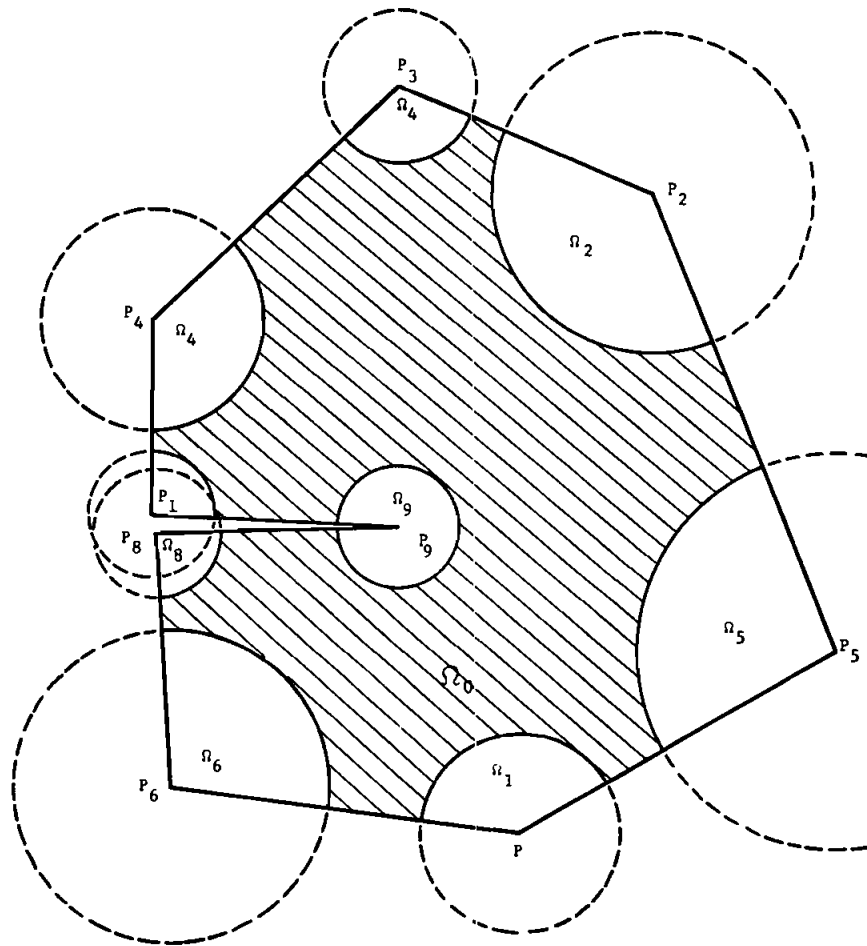


Fig. 3 Localization of domains.

outside $\hat{\Omega}$. Schatz and Whalbin [6] have shown that for $s > \beta_M - 1$,

$$\|u - u_h\|_{-s, \Omega} \leq C(u) h^{\min(2k, 2\beta_M) - \epsilon} \quad (23)$$

Hence, upon combining (21), (22), and (23), we find that

$$\|u - u_h\|_{\infty, \Omega_0} \leq C(u) h^{\min(k+1, 2\beta_M) - \epsilon} \quad (24)$$

Clearly, whenever $1/2 < \beta_M < 1$, which is the case in which Ω is a cracked or L-shaped domain, the local interior error in $L^\infty(\Omega_0)$ is of order $h^{2\beta_M - \epsilon}$. For example, for the domains in Fig. 2, we have

$$\begin{aligned} \text{Rectangular: } \|u - u_h\|_{\infty, \Omega_0} &\leq C(u) h^{\min(k+1, 4) - \epsilon} \\ \text{L-shaped: } \|u - u_h\|_{\infty, \Omega_0} &\leq C(u) h^{4/3 - \epsilon} \\ \text{Cracked: } \|u - u_h\|_{\infty, \Omega_0} &\leq C(u) h^{1 - \epsilon} \end{aligned} \quad (25)$$

The dominating factor here, of course, is the so-called *pollution* of the local behavior due to irregularities in the solution outside of the test domain Ω_0 . No matter how rich the local interpolation functions (i.e., no matter how high the degree of the polynomials used in the finite element approximation), the local rates of convergence are the same for the L-shaped and cracked domains.

Schatz and Whalbin [6] have also obtained the rate of convergence at a point x a distance d from a vertex. Suppose $\beta_M < 1$ (e.g., cracked or L-shaped domains) and that regular or quasi-uniform mesh refinements are used. Then

$$|(u - u_h)(x)| \leq \begin{cases} C(u) h^{2\beta_M - \epsilon} d^{-\beta_M}, & h \leq d \leq d_0 \\ C(u) h^{\beta_M - \epsilon} & d \leq h \end{cases} \quad (26)$$

In other words, a substantial increase in the rate of convergence is experienced as one goes from the element nearest the vertex (or crack) to adjacent elements a distance $d \leq h$ from the vertex.

IV. STRESS INTENSITY FACTORS

Let us again consider the model problem

$$\begin{aligned} -\Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \end{aligned} \quad (27)$$

where Ω is a polygonal domain of the type shown in Fig. 1. Suppose that $f \in H^s(\Omega)$ where generally $s = 0$ but s is not a number of the set of exceptional values $\{(n\pi/\alpha_j) - 1, n = 1, 2, \dots\}$. Then the solution u of (27) is of the form

$$u(\mathbf{x}) = \sum_{m=1}^N k_m \psi_m(\mathbf{x}) + w(\mathbf{x}) \quad (28)$$

where $\mathbf{x} = (x, y) \in \Omega \subset \mathbf{R}^2$, N is the number of vertices for which $\alpha_j > \pi$, k_m are constants, $\psi_m(\mathbf{x})$ are the so-called *singular functions*, and $w(\mathbf{x})$ is a "smooth part" of the solution in the sense that w satisfies the regularity estimate

$$\|w\|_{s+2, \Omega} \leq C \|f\|_{s, \Omega} \quad (29)$$

Next, consider a typical vertex P_j of the type shown in Fig. 4 and let Ω_j denote the intersection of Ω with an open ball of radius R_j centered at P_j , and Ω_j^o the intersection of Ω with a ball of radius $R_j^o < R_j$ centered at P_j . As indicated in the figure, we establish a polar coordinate system (r, θ) at P_j , $0 \leq \theta \leq \alpha_j$. Then, in the neighborhood of this wedge, the solution of (27)

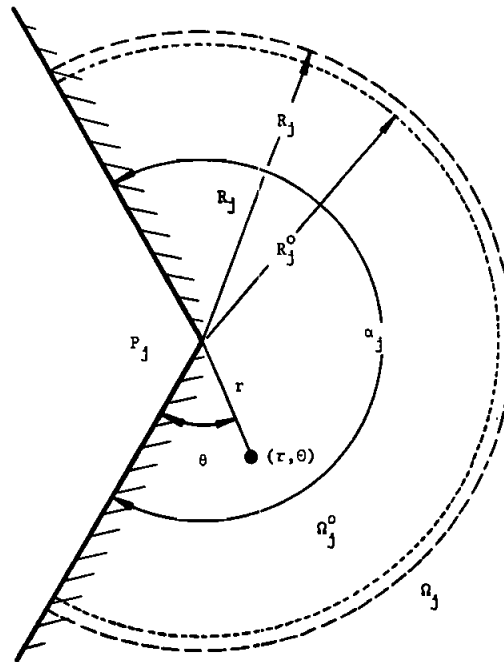


Fig. 4 Polar coordinates in a neighborhood of vertex P_j .

assumes the form

$$u(r, \theta) = w + \sum_{m \in I} k_m^j \psi_m^j(r, \theta) \quad (30)$$

where

$$I = \{m: m \text{ integral}, 0 < m < (s+1)\alpha_j/\pi\}$$

and

$$\begin{aligned} \psi_m^j(r, \theta) &= r^{m\beta_j} \omega_j(r) \sin \frac{\pi\theta}{\alpha_j} \\ \omega_j &\in C^\infty(\bar{\Omega}): \omega_j(r) = \begin{cases} 1, & 0 \leq r \leq R_j^0 \\ 0, & r \geq R_j \end{cases} \end{aligned} \quad (31)$$

The coefficients k_m^j satisfy

$$|k_m^j| \leq C \|f\|_{s, \Omega} \quad (32)$$

The derivatives of the singular functions ψ_m^j have the growth property

$$|D^\alpha \psi_m^j(r, \theta)| \leq C r^{m\beta_j - |\alpha| - \epsilon} \quad (33)$$

Here $\alpha = (\alpha_1, \alpha_2)$; α_i integral ≥ 0 ,

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x^{\alpha_1} \partial y^{\alpha_2}}, \quad |\alpha| = \alpha_1 + \alpha_2$$

To simplify the discussion, consider the cracked domain shown in Fig. 5. In this case (with obvious notations),

$$u(r, \theta) = w + k\omega(r)r^{1/2} \sin(\theta/2) \quad (34)$$

or

$$u(r, \theta) = w + k\phi(r, \theta) \quad (34a)$$

It is an easy matter to show that in Eq. (34) $u \in H^{3/2-\epsilon}(\Omega)$, as one would expect (see Eq. 17). The constant k in Eq. (34) is a measure of the strength of the singularity and is called the stress intensity factor, borrowing from

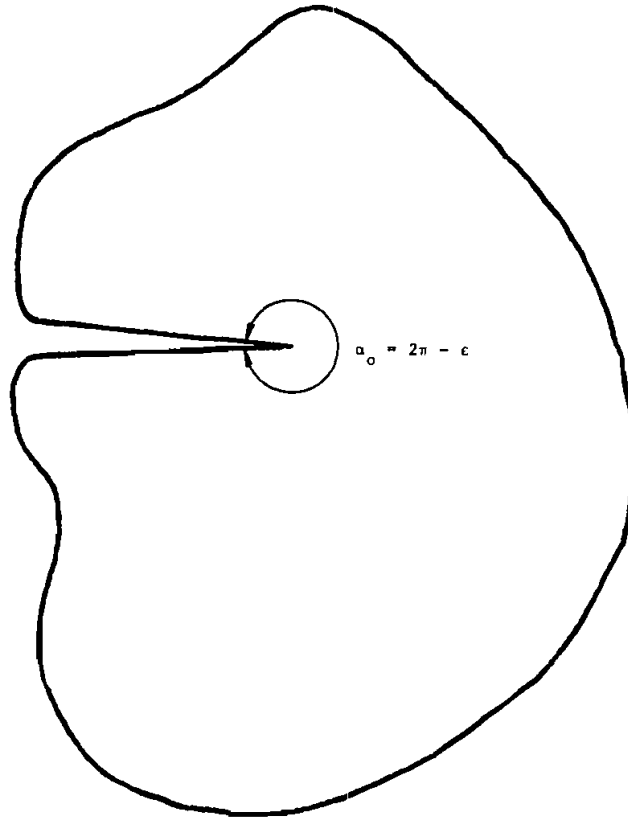


Fig. 5 Smooth domain with a crack.

elastic fracture mechanics. In linear elastic fracture mechanics applications a finite element analysis can be performed from which values of the stress intensity factor k are calculated. These calculated values for k can then be compared to the critical value k^* , and if k is less than k^* the crack is stable. On the other hand, if k is larger than k^* the crack is unstable and propagation will begin.

There are numerous techniques available [2] for the calculation of k from a finite element solution. One of the simplest stems from the observation that

$$k = \lim_{d \rightarrow 0} d^{-1/2} [u(d, \theta) - u(0, \theta)] \quad (35)$$

where d is a measure of distance from the crack vertex and $0 \leq \theta \leq \pi$. Schatz and Whalbin [6] have employed this concept to arrive at the following

approximation for the stress intensity factor k_h :

$$k_h = \frac{u_h(d, \theta)}{d^{1/2} \sin\left(\frac{\theta}{2}\right)} \quad (36)$$

They suggest that k_h in (36) is best evaluated at $d = h^{2/3}$ and obtain the following *a priori* error estimate:

$$|k - k_h| \leq Ch^{1/3-\epsilon} \quad (37)$$

V. A HYBRID ELEMENT

A great deal has been claimed in the literature about the advantages of hybrid elements for solving crack problems in elasticity (see, for example, the survey article of Pian [2] or the work of Atluri [11] or Tong [12]). Mathematical theories for hybrid elements for smooth domains have been developed by Babuska, Oden, and Lee [14, 15].

We describe here one representative type of hybrid finite element, the analysis of which uses ideas similar to those used in the application of Lagrange multipliers in finite element methods presented by Babuska [17].

In order to demonstrate this procedure we will consider the model problem.

$$\begin{aligned} -\Delta u + u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \end{aligned} \quad (38)$$

wherein Ω is the domain shown in Fig. 6. Let

$$\begin{aligned} \mathcal{U}_0(\Omega_0) &= \{u \in H^1(\Omega_0) : u = 0 \text{ on } \partial\Omega_0 - \Gamma\} \\ \mathcal{U}_1(\Omega_1) &= \{v \in H^1(\Omega_1) : v = 0 \text{ on } \partial\Omega_1 - \Gamma\} \\ \mathcal{M}(\Gamma) &= \{\psi \in L_2(\Gamma) : \exists \sigma \in \mathcal{S} \ni \sigma \cdot \mathbf{n} = \psi \text{ on } \Gamma\} \\ \mathcal{S} &= \{\sigma \in L_2(\Omega) : \nabla \cdot \sigma \in L_2(\Omega_\alpha), \alpha = 1, 2\} \end{aligned} \quad (39)$$

We can then define the product space

$$\mathcal{X} = \mathcal{U}_0(\Omega_0) \times \mathcal{U}_1(\Omega_1) \times \mathcal{M}(\Gamma) \quad (40)$$

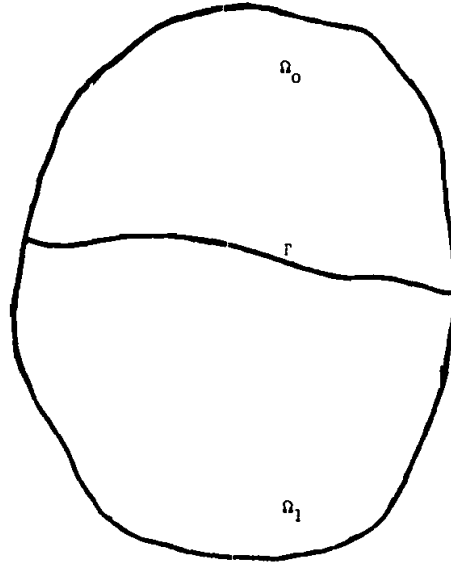


Fig. 6 Two-domain model.

with the norm

$$\begin{aligned} \|\lambda\|_{\mathcal{X}}^2 &= \|u\|_{1,\Omega_0}^2 + \|v\|_{1,\Omega_1}^2 + \|\psi\|_{-1/2,\Gamma}^2 \\ \lambda &= (u, v, \psi) \in \mathcal{X} \end{aligned} \quad (41)$$

Over the above defined space $\mathcal{X}$ we can define the following bilinear form and linear functional:

$$\begin{aligned} B(\lambda, \bar{\lambda}) &= \int_{\Omega_0} (\nabla u \cdot \nabla \bar{u} + u \bar{u}) dx + \int_{\Omega_1} (\nabla v \cdot \nabla \bar{v} + v \bar{v}) dx \\ &\quad + \int_{\Gamma} [\bar{\psi}(u + v) + \psi(\bar{u} + \bar{v})] ds \\ F(\bar{\lambda}) &= \int_{\Omega_0} f \bar{u} dx + \int_{\Omega_1} f \bar{v} dx \end{aligned} \quad (42)$$

We now consider the variational boundary value problem of finding $\lambda \in \mathcal{X}$ such that

$$B(\lambda, \bar{\lambda}) = F(\bar{\lambda}), \quad \forall \bar{\lambda} \in \mathcal{X} \quad (43)$$

Without much difficulty it can be seen that this problem is equivalent to

$$\begin{aligned} -\Delta u + u &= f \text{ in } \Omega_0 \\ -\Delta v + v &= f \text{ in } \Omega_1 \\ \frac{\partial u}{\partial n} + \psi &= 0, \quad \frac{-\partial v}{\partial n} + \psi = 0, \quad u - v = 0 \text{ on } \Gamma \end{aligned} \quad (44)$$

which in turn is equivalent to problem (38). The demonstration of continuity of the bilinear form is a rather straightforward procedure leading to

$$|B(\lambda, \hat{\lambda})| \leq 2\|\lambda\|_x \|\hat{\lambda}\|_x \quad (45)$$

In order to investigate the inf sup condition on $B(\cdot, \cdot)$ we first need to consider the auxiliary problem

$$\begin{aligned} -\Delta z_\alpha + z_\alpha &= 0 \text{ in } \Omega_\alpha, \quad \alpha = 0, 1 \\ \partial z_\alpha / \partial n &= \psi \text{ on } \Gamma \end{aligned} \quad (46)$$

One obtains

$$\begin{aligned} \int_\Gamma \psi u \, ds &= (z_0, u)_{1, \Omega_0} \\ \int_\Gamma \psi v \, dx &= (z_1, v)_{1, \Omega_1} \\ \|\psi\|_{\mathcal{A}}^2 &= \|\psi\|_{-1/2, \Gamma}^2 = \|z_\alpha\|_{1, \Omega_\alpha}^2, \quad \alpha = 0, 1 \end{aligned} \quad (47)$$

By letting

$$\begin{aligned} \hat{v} &= v + z_1 \\ \hat{u} &= u + z_0 \\ \hat{\psi} &= -2\psi \end{aligned}$$

we obtain

$$|B(\lambda, \hat{\lambda})| \geq \|\lambda\|_x^2$$

and

$$\|\hat{\lambda}\|_x \leq \sqrt{8}\|\lambda\|_x$$

which yields

$$\inf_{\|\lambda\|_{\mathcal{X}}=1} \sup_{\|\bar{\lambda}\|_{\mathcal{X}} \leq 1} |B(\lambda, \bar{\lambda})| \geq \frac{1}{\sqrt{8}} \quad (48)$$

Hence a unique solution to problem (43).

In order to discuss the finite element approximation for problem (43), we need to introduce the following finite dimensional subspaces:

- $\mathcal{Q}_k(\Omega_1)$ = space of usual piecewise polynomials of degree $\leq k$ satisfying the homogeneous boundary condition on $\partial\Omega_1 - \Gamma$
- $\mathcal{Q}_r(\Omega_0)$ = space of usual piecewise polynomials of degree $\leq r$ on Ω_0 satisfying the homogeneous boundary condition on $\partial\Omega_0 - \Gamma$
- $\mathcal{Q}_t(\Gamma)$ = space of piecewise polynomials Ψ of degree $\leq t$ on Γ such that $\Psi \in \mathcal{M}(\Gamma)$

We set

$$\mathcal{X}_h = \mathcal{Q}_r(\Omega_0) \times \mathcal{Q}_k(\Omega_1) \times \mathcal{Q}_t(\Gamma) \subset \mathcal{X}$$

Then the finite element approximation associated with problem (43) is to find $\Lambda \in \mathcal{X}_h$ such that

$$B(\Lambda, \bar{\Lambda}) = F(\bar{\Lambda}) \quad \forall \bar{\Lambda} \in \mathcal{X}_h \quad (49)$$

Again, it is an easy matter to show that

$$|B(\Lambda, \bar{\Lambda})| \leq 2\|\Lambda\|_{\mathcal{X}}\|\bar{\Lambda}\|_{\mathcal{X}} \quad (50)$$

To show the inf sup condition let

$$\begin{aligned} \vartheta &= V + \Pi_1 z_1 \\ \hat{U} &= U + \Pi_0 z_0 \\ \hat{\Psi} &= -2\Psi' \end{aligned} \quad (51)$$

where Π_1 and Π_0 denote the orthogonal projections

$$\begin{aligned} \Pi_1 &: H^1(\Omega_1) \rightarrow \mathcal{Q}_k(\Omega_1) \\ \Pi_0 &: H^1(\Omega_0) \rightarrow \mathcal{Q}_r(\Omega_0) \end{aligned} \quad (52)$$

We then obtain

$$\inf_{\|\Lambda\|_X=1} \sup_{\|\bar{\Lambda}\|_X \leq 1} |B(\Lambda, \bar{\Lambda})| \geq \frac{\beta}{\sqrt{8}} \quad (53)$$

where

$$\begin{aligned} \beta &= \min \{1, \mu_0, \mu_1\} \\ \mu_0 &= \inf_{\Psi \in Q_r} \frac{\|\Pi_0 z_0\|_{1, \Omega_0}}{\|\Psi\|_{-1/2, \Gamma}^2} \\ \mu_1 &= \inf_{\Psi \in Q_r} \frac{\|\Pi_1 z_1\|_{1, \Omega_1}^2}{\|\Psi\|_{-1/2, \Gamma}} \end{aligned} \quad (54)$$

Hence a unique solution to the finite element approximation exists provided μ_0 and μ_1 are strictly positive and independent of any mesh parameter.

Following the arguments of Ref. 16, we see that μ_0 and μ_1 are strictly positive if, and only if, for any $\Psi \in Q_r(\Gamma)$,

$$\begin{aligned} \int_{\Gamma} \Psi U \, dx &= 0 \quad \forall U \in Q_r(\Omega_0) \\ \int_{\Gamma} \Psi V \, dx &= 0 \quad \forall V \in Q_r(\Omega_1) \end{aligned} \quad \text{implies that } \Psi = 0 \quad (55)$$

Equation (55) is referred to as the *rank condition*. If we let

$$\begin{aligned} h_0 &= \text{mesh parameter for } \Omega_0 \\ h_1 &= \text{mesh parameter for } \Omega_1 \\ h_{\Gamma} &= \text{mesh parameter for } \Gamma \end{aligned} \quad (56)$$

Then, following the logic of Ref. 17, we see that condition (55) is equivalent to the existence of two positive constants M_0, M_1 such that

$$h_{\Gamma}/h_0 \geq M_0, \quad h_{\Gamma}/h_1 \geq M_1 \quad (57)$$

In order to obtain estimates of the error associated with the above method, we first recall the approximation estimate (12) which states

$$\|\mathbf{e}\|_X \leq A_0 \inf_{\bar{\Lambda} \in \bar{X}_h} \|\lambda - \bar{\Lambda}\|_X \quad (58)$$

wherein

$$A_0 = 1 + \frac{\sqrt{8}}{\beta}$$

and β is defined in (54). The previously defined finite element subspaces are

endowed with the following standard interpolation properties:

For any $u \in H^l(\Omega_0) \exists C_1 > 0$ and $\tilde{U} \in Q_r(\Omega_0)$ such that

$$\begin{aligned} \|u - \tilde{U}\|_{1,\Omega_0} &\leq C_1 h_0^\mu \|u\|_{l,\Omega_0} \\ \mu &= \min \{r, l-1\} \end{aligned} \quad (59)$$

For any $v \in H^l(\Omega_1) \exists C_2 > 0$ and $\tilde{V} \in Q_r(\Omega_1)$ such that

$$\begin{aligned} \|v - \tilde{V}\|_{1,\Omega_1} &\leq C_2 h_1^\omega \|v\|_{l,\Omega_1} \\ \omega &= \min \{k, l-1\} \end{aligned} \quad (60)$$

For any $\psi \in H^{-1/2}(\Gamma) \exists C_3 > 0$ and $\tilde{\Psi} \in Q_t(\Gamma)$ such that

$$\begin{aligned} \|\psi - \tilde{\Psi}\|_{-1/2,\Gamma} &\leq C_3 h_\Gamma^\theta \|u\|_{l,\Omega_0} \\ \theta &= \min \{t+1, l-1\} \end{aligned} \quad (61)$$

Combining Eqs. (58) and (59)–(61), we obtain

$$\|e\|_x \leq A_0 C h^\alpha \|u\|_{l,\Omega_0} \quad (62)$$

where

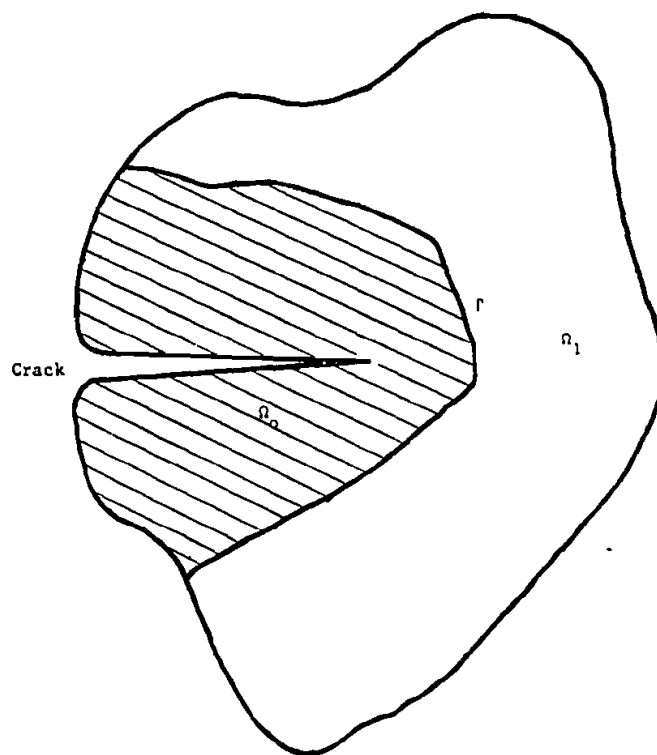
$$\begin{aligned} h &= \max \{h_0, h_1, h_\Gamma\} \\ \alpha &= \min \{k, r, t+1, l-1\} \end{aligned} \quad (63)$$

In order to consider the effect of using a special crack element, we will consider the same variational principle as above, but the domain will contain a crack in Ω_0 (see Fig. 7). Suppose that on Ω_1 and Γ we define the same finite element subspaces $Q_k(\Omega_1)$ and $Q_t(\Gamma)$, but now we will let Ω_0 be a single element (h_0 now being the diameter of Ω_0), and we will define over Ω_0 a new approximation space $Q_r^s(\Omega_0)$ where

$$Q_r^s(\Omega_0) = \{U \in H^1(\Omega_0): U = W + K\phi; W \in Q_r(\Omega_0)\} \quad (64)$$

in which K is a constant which will be the finite element approximation to k in Eq. (27), and ϕ is the exact form of the singular function found in Eq. (34a) (i.e., $\phi = \omega(r)r^{1/2} \sin \theta/2$). We see that so long as $h_0 \geq 2R^0$ (R^0 defined in Eq. 31), the singularity is defined over only element Ω_0 . Hence, having imbedded the space $Q_r^s(\Omega_0)$ with singular functions of the type occurring in the actual problem, the logical question is what order of estimate is to be expected over Ω_0 :

$$\begin{aligned} \inf_{U \in Q_r^s(\Omega_0)} \|u - U\|_{1,\Omega_0} &= \inf_{U \in Q_r^s(\Omega_0)} \|w + k\phi - W - K\phi\|_{1,\Omega_0} \\ &\leq \inf_{W \in Q_r^s(\Omega_0)} \|w - W\|_{1,\Omega_0} + \inf_{K \in \mathbb{R}} |k - K| \|\phi\|_{1,\Omega_0} \end{aligned} \quad (65)$$

Fig. 7 Crack element Ω_0 .

We see that the second term in Eq. (65) vanishes, leaving us with

$$\begin{aligned} \inf_{U \in Q_r^s(\Omega_0)} \|u - U\|_{\Omega_0} &\leq \inf_{W \in Q_r(\Omega_0)} \|w - W\|_{1, \Omega_0} \\ &\leq C_3 h_0^\eta \|w\|_{l, \Omega_0}, \quad \eta = \min \{r, l - 1\} \end{aligned} \quad (66)$$

which leads to the estimate Eq. (62).

A word of caution needs to be expressed: Since our development was based on the assumption that $h_0 \geq 2R^0$, we can no longer allow h_0 to become "small" but rather h_0 is a fixed predetermined value. This property of our results can be overcome by allowing Ω_0 to contain more than a single element; i.e., by letting $Q_r^s(\Omega_0)$ become a hybrid approximation space with mesh parameter h_0 and enforcing upon Ω_0 the condition that $\text{dia}(\Omega_0) \geq 2R_0$.

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